CRM Orchestrator — Load Test & Capacity Estimates

Empirical Benchmarks (Local Machine) vs. Cloud Free-Tier Deployment (Vercel & Railway)

1. Executive Summary

This report evaluates the throughput, message capacity, and latency limits of the CRM Orchestrator system under three hosting environments: **(1) Local Workstation** (Intel i7-13650HX, 20 vCPUs, 24GB RAM), **(2) Vercel Serverless Hosting** (Free Hobby Tier), and **(3) Railway PaaS** (Free Trial / Hobby Container Tier). The evaluation separates the fast path (HTTP routing, compliance verification, and queue dispatch) from the heavy path (LLM draft generation).

2. Environment Specifications & Constraints

Parameter	Local Machine (Tested)	Railway (Free / Starter)	Vercel (Hobby Free Tier)
Compute / CPU	20 Cores (Intel i7-13650HX)	Shared 0.5–1.0 vCPU	1 vCPU (Serverless Function)
RAM / Memory	24 GB DDR5	512 MB – 1 GB capped	1024 MB per execution
Execution Model	Persistent process (Uvicorn)	Persistent Docker container	Serverless Lambda (Ephemeral)
Max Timeout	Unlimited	Unlimited (Persistent)	15 seconds max execution
Local LLM / BERT	Supported (Ollama + DistilBERT)	No GPU; OOM risk for 7B models	Impossible (No persistent state)
Database	Local SQLite3 WAL mode	SQLite volume / Postgres	External only (Supabase/Neon)

3. Empirical Load Test Results (Measured Locally)

Endpoint / Subsystem	Median (p50)	p95 Latency	Max Measured Throughput	Operational Bottleneck
FastAPI Read Routing GET /api/health	6.36 ms	13.09 ms	**~940 – 1,050 req/sec**	Network stack / asyncio loop
Session & State Mgmt GET /api/chat/sessions	5.67 ms	7.42 ms	**~980 req/sec**	SQLite concurrent read locks
Compliance Cascade Stage 1 (Regex) + Stage 2 (BERT)	1.21 ms	2.45 ms	**823.7 evals/sec** (~49k/min)	CPU vector string calculations
Dispatch Engine (Preflight) POST /api/dispatch/preflight	0.86 ms	1.41 ms	**1,176 checks/sec**	In-memory regex & schema audit
Mock Aggregator Dispatch Outgoing HTTP dispatch loop	50.0 ms	52.1 ms	**20 msgs/sec** (1,200 msgs/min)	Configured send_interval_ms=50
LLM Generation (Ollama) qwen2.5:7b-instruct draft	1,380 ms	1,850 ms	**0.7 – 1.0 drafts/sec** (CPU)	Ollama single-stream CPU inference

4. Message Sending & Pipeline Capacity Estimates

Capacity varies significantly depending on whether the workload is **Outbound SMS/WhatsApp Dispatch** (evaluating templates & sending to Aggregator API) or **Conversational Chat & Draft Generation** (running the LLM agent).

Metric / Workload Tier	Local Machine (Tested)	Railway (Free Tier)	Vercel (Hobby Tier)
Compliance Screening RPS	**820+ msgs/sec**	~150–250 msgs/sec	~200–350 msgs/sec
Outbound Dispatch Rate	**20 msgs/sec** (1,200/min) *(Capped by 50ms interval)*	**20 msgs/sec** (1,200/min) *(Up to ~50/s unthrottled)*	**5–10 msgs/sec** *(Serverless connection limits)*
Max Messages Per Single Batch	100,000+ msgs (persistent)	10,000–25,000 msgs / batch	Max ~200 msgs per invocation *(15-second execution timeout)*
Monthly Message Ceiling	**Millions** (Hardware bound only)	**~500,000 / month** *(credit quota ceiling)*	**~100,000 / month** *(100k edge invocations limit)*
LLM Generation Throughput	**~1 draft / 1.4s** (~42 drafts/min) *(Local Ollama CPU)*	**External API Required** (Groq / Claude / OpenAI)	**External API Required** (Groq / Claude / OpenAI)
Max Concurrent Active Users	20–30 simultaneous sessions	5–15 simultaneous sessions	50+ concurrent requests (burst)

5. Production Recommendations & Architecture Strategy

Recommended Setup A: Hybrid Architecture (Recommended & Most Cost-Effective)

- **Mock Aggregator on Vercel:** Perfectly suited for Vercel. Because the mock aggregator simply accepts HTTP POST webhooks and responds with message SIDs, it operates well within Vercel's 15-second limit and zero-cost free hobby quota.
- **CRM Orchestrator on Railway or Local Server:** Running the CRM backend on Railway (or your local machine via Cloudflare Tunnel / ngrok) allows persistent SQLite/Postgres storage, unhindered SSE streaming, and batch job dispatching without serverless timeout cutoffs.

Recommended Setup B: Pure Cloud Deployment (Free Tiers)

- **Switch LLM to Groq / External API:** On Railway Free Tier (512MB RAM), self-hosting a 7B LLM is impossible. Configure config/app.yaml to point to Groq's free-tier API (Llama 3.3 / Qwen) which generates drafts at **250+ tokens/sec** (sub-second latency) with zero RAM impact.
- **Externalize SQLite to Supabase/Neon:** If hosting the CRM on Vercel, replace local SQLite with a serverless PostgreSQL database (e.g., free tier Neon or Supabase) to avoid ephemeral container storage resets on each deployment.

Key Bottleneck Safeguards:

1. **Dispatch Rate Limit:** send_interval_ms: 50 guarantees deterministic 20 msgs/sec delivery, preventing aggregator rate limits.
2. **Compliance Cascade Efficiency:** Stage 1 Regex + Stage 2 Classifier evaluates at **823 msgs/sec**, meaning compliance checks will never bottle up dispatch.
3. **SQLite Concurrency:** Enabled with WAL (Write-Ahead Logging) mode, sustaining up to 1,000 read req/sec locally before lock contention.